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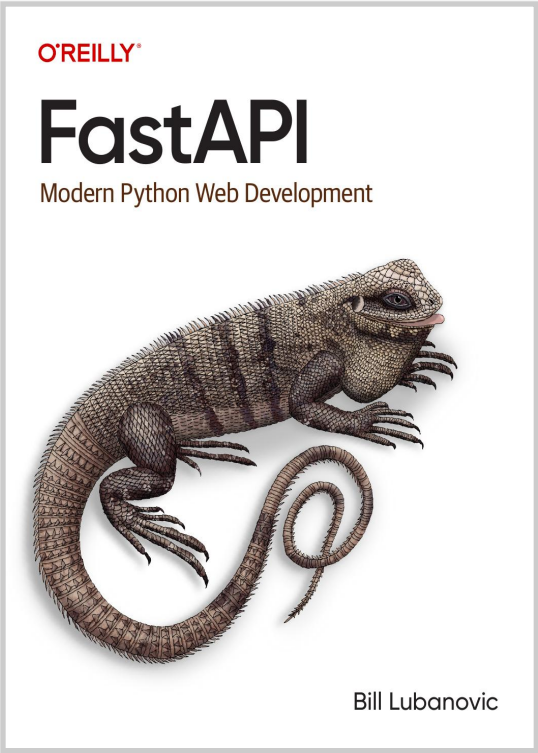

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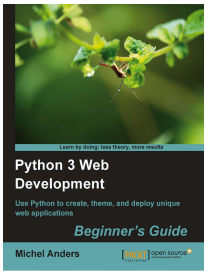

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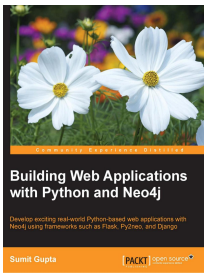

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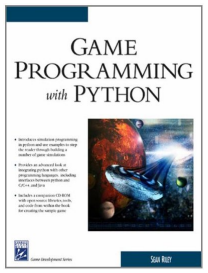

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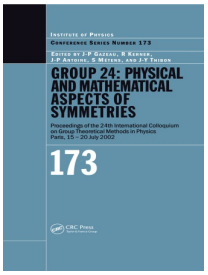

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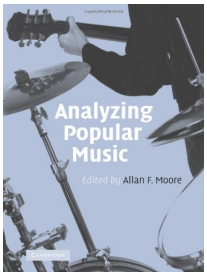

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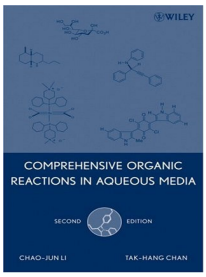

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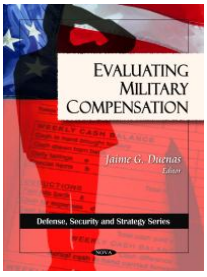

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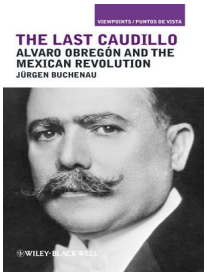

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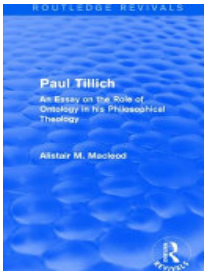

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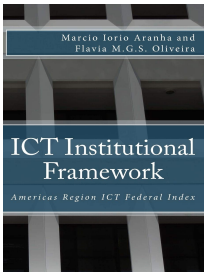

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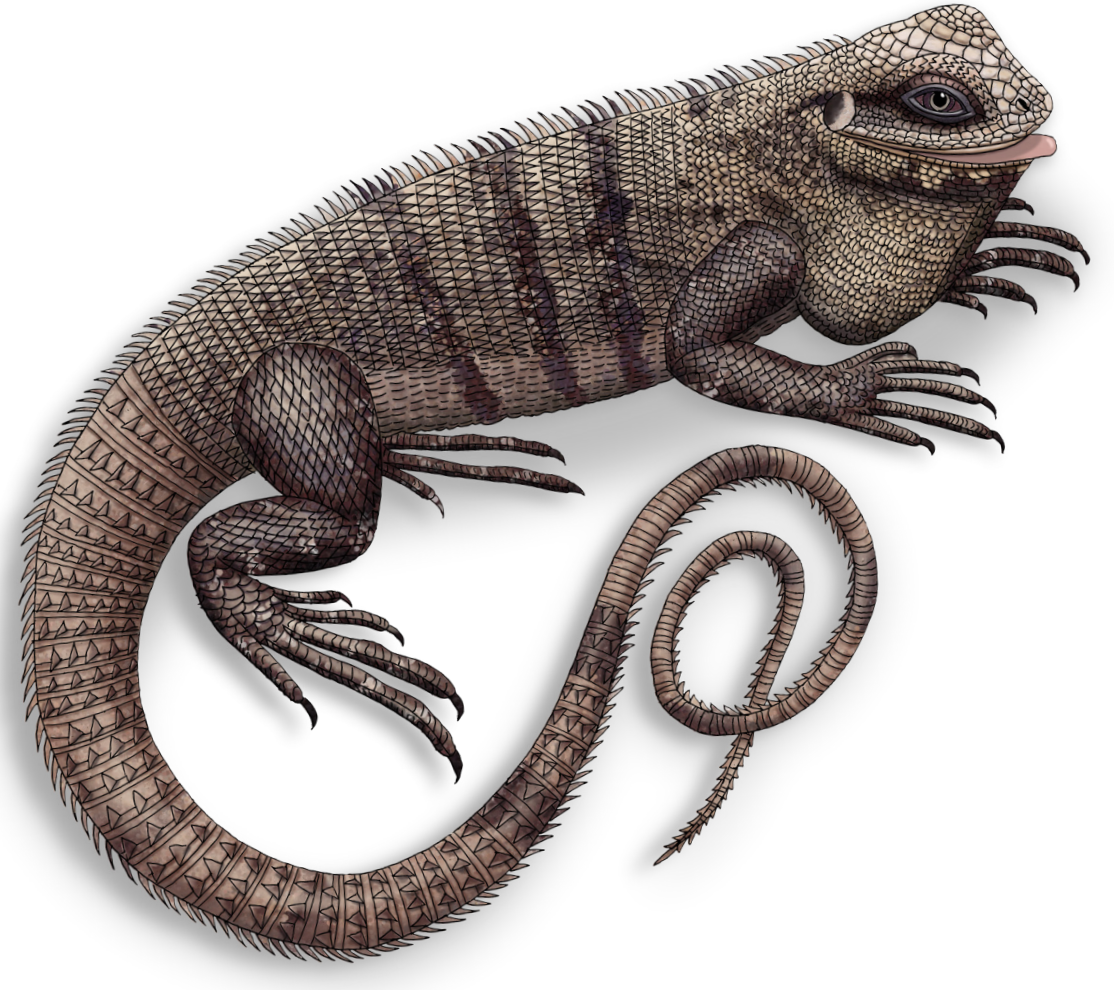

Bill Lubanovic

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FastAPI

FastAPI is a young yet solid framework that takes advantage of newer Python features in a clean design. As its name implies, FastAPI is indeed fast, rivaling similar frameworks in languages such as Golang. With this practical book, developers familiar with Python will learn how FastAPI lets you accomplish more in less time with less code.

Author Bill Lubanovic covers the nuts and bolts of FastAPI development with how-to guides on various topics such as forms, database access, graphics, maps, and more that will take you beyond the basics. The book also gets you up to speed on RESTful APIs, data validation, authorization, and performance. With its similarities to frameworks like Flask and Django, you'll find it easy to get started with FastAPI.

Through the course of this book, you will:

- **Learn how to build web applications with FastAPI**
- **Understand the differences between FastAPI, Starlette, and Pydantic**
- **Learn two features that set FastAPI apart: asynchronous functions and data type checking and validation**
- **Examine new features of Python 3.8+, especially type annotations**
- **Understand the differences between sync and async Python**
- **Learn how to connect with external APIs and services**

Bill Lubanovic has been a developer for over 40 years, specializing in Linux, the Web, and Python. He recently used FastAPI with his team to rewrite a large biomedical research API. Bill coauthored *Linux System Administration* and wrote *Introducing Python*, both for O'Reilly.

"FastAPI made simple! This book excels at simplifying FastAPI concepts, showcasing the author's mastery. Readers will gain practical knowledge and hit the ground running."

—**Ganesh Harke**
Senior Software Engineer, Citibank

"This book provides a comprehensive overview of the FastAPI framework and its surrounding ecosystem, giving readers a quick yet comprehensive view of modern web development."

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by Bill Lubanovic

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[LSI]

To the loving memory of my wife, Mary, my parents, Bill and Tillie, and my friend, Rich.
I miss you.

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Preface

This is a pragmatic introduction to FastAPI—a modern Python web framework. It's also a story of how, now and then, the bright and shiny objects that we stumble across can turn out to be very useful. A silver bullet is nice to have when you encounter a werewolf. (And you will encounter werewolves later in this book.)

I started programming scientific applications in the mid-1970s. And after I first met Unix and C on a PDP-11 in 1977, I had a feeling that this Unix thing might catch on.

In the '80s and early '90s, the internet was still noncommercial, but already a good source for free software and technical info. And when a web browser called Mosaic was distributed on the baby open internet in 1993, I had a feeling that this web thing might catch on.

When I started my own web development company a few years later, my tools were the usual suspects at the time: PHP, HTML, and Perl. On a contract job a few years later, I finally experimented with Python and was surprised at how quickly I was able to access, manipulate, and display data. In my spare time over two weeks, I was able to replicate most of a C application that had taken four developers a year to write. Now I had a feeling that this Python thing might catch on.

After that, most of my work involved Python and its web frameworks, mostly Flask and Django. I particularly liked the simplicity of Flask and preferred it for many jobs. But just a few years ago, I spied something glinting in the underbrush: a new Python web framework called FastAPI, written by Sebastián Ramírez.

As I read his (excellent) documentation, I was impressed by the design and thought that had gone into it. In particular, his history page showed how much care he had taken evaluating alternatives. This was not an ego project or a fun experiment, but a serious framework for real-world development. Now I had a feeling that this FastAPI thing might catch on.

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